

### 1.3 Classwork: Series and Sigma Notation

1. An arithmetic series has first term  $u_1 = 3$  and common difference  $d = 5$ .

(a) Find  $u_{20}$ , the twentieth term. [2]

(b) Find  $S_{20}$ , the sum of the first 20 terms. [2]

(c) The  $n^{th}$  term of the sequence is 148. Find  $n$ . [2]

2. A geometric sequence has first term  $u_1 = 6$  and common ratio  $r = \frac{1}{2}$ .

(a) Write down the first four terms of the sequence. [2]

(b) Find  $S_8$ , the sum of the first 8 terms. [2]

(c) Explain why the infinite sum  $S_\infty$  exists, and find its value. [3]

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3. Consider the arithmetic sequence  $7, 11, 15, 19, \dots$

(a) Find the general term  $u_n$ . [2]

(b) Find  $\sum_{k=1}^{15} u_k$ . [2]

(c) Find the least value of  $n$  such that  $S_n > 1000$ . [3]

4. The sum of the first  $n$  terms of a geometric series is given by

$$S_n = 4 \left( 1 - \left( \frac{1}{3} \right)^n \right)$$

- (a) Find  $S_1$  and hence write down  $u_1$ . [2]
- (b) Find  $S_2$  and hence find  $u_2$ . [2]
- (c) Find the common ratio  $r$ . [1]
- (d) Find  $S_\infty$ . [2]